

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
CONCEPT RECAPITULATION TEST – II
PAPER –1
TEST DATE: 30-06-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. C
2. A
3. B
4. C

5. AD

Sol. By the energy conservation (ME) between bottom point and point Y

$$\frac{1}{2}mv_0^2 = mgh + \frac{1}{2}mv_1^2$$

$$\therefore v_1^2 = v_0^2 - 2gh \quad \dots(i)$$

Now at point Y the centripetal force provided by the component of mg

$$\therefore mg \sin 30^\circ = mv_1^2 / R$$

$$\therefore v_1^2 = gR / 2$$

From (i)

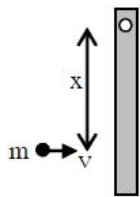
$$\frac{gR}{2} = v_0^2 - 2gh$$

At point x and z of circular path, the points are at same height but less than h. So the velocity more than a point y

So required centripetal $= mv^2 / r$ is more.

6. ACD

Sol. By the angular momentum conservation about the suspension point



$$mvx = \left(\frac{m\ell^2}{3} + mx^2 \right) \omega$$

$$\therefore \omega = \frac{mvx}{m\ell^2/3 + mx^2} = \frac{2vx}{\ell^2 + 3x}$$

$$\text{For maximum } \therefore \omega \Rightarrow \frac{d\omega}{dx} = 0$$

$$\therefore X_M = \ell / \sqrt{3}$$

$$\text{So the } \omega = V / 2\ell\sqrt{3}$$

7. AC

Sol. $\lambda_{\min} = \frac{hv}{eV}$

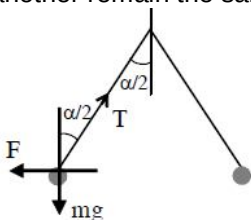
$$\Rightarrow \lambda_{\min} \propto \frac{1}{V} \Rightarrow (\lambda_{\min})_{\text{new}} = \frac{\lambda_2}{2}$$

$$\therefore I = \frac{dN}{dt} \times \frac{hc}{\lambda}$$

$$\therefore \frac{dN}{dt} \text{ decreases}$$

8. AC

Sol. The net electric force on any sphere is lesser but by Coulomb law the force due to one sphere to another remain the same.



In equilibrium $T \cos \alpha/2 = mg$ and $T \sin \alpha/2 = F$

After immersed in dielectric liquid. As given no change in angle α .

So $T \cos \alpha/2 = mg - V\rho g$ when $\rho = 800 \text{ Kg/m}^3$ and $T \sin \alpha/2 = F / e_r$

$$\therefore \frac{mg}{F} = \frac{mg - V\rho g}{F / e_r}$$

d = density of sphere

$$\frac{1}{21} = 1 - \frac{800}{d} \Rightarrow d = 840$$

9. ABCD

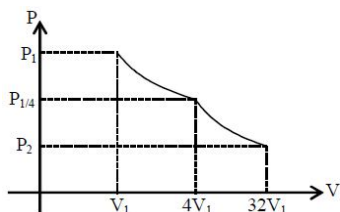
10. AC

Section – B

11. 9

Sol. $\frac{P_1}{4}(4V_1)^{5/3} = P_2(32V_1)^{5/3}$

$$P_2 = \frac{P_1}{4}(1/8)^{5/3} = \frac{P_1}{128}$$



$$W_{\text{adi}} = \frac{P_1V_1 - P_2V_2}{\gamma - 1} = \frac{P_1V_1 - \frac{P_1}{128}(32V_1)}{5/3 - 1} = \frac{P_1V_1(3/4)}{2/3} = \frac{9}{8}P_1V_1$$

$$W_{\text{iso}} = P_1V_1 \ln\left(\frac{4V_1}{V_1}\right) = 2P_1V_1 \ln 2$$

$$\frac{W_{\text{iso}}}{W_{\text{adi}}} = \frac{2P_1V_1 \ln 2}{9/8 P_1V_1} = \frac{16}{9} \ln 2$$

12. 5

Sol. $f \propto \frac{1}{\ell_1} \Rightarrow f = \frac{c}{\ell_1} \dots\dots\dots(1)$

($\ell_1 \Rightarrow$ initial length of pipe)

$$\left(\frac{V}{V - V_T}\right)f = \frac{c}{\ell_2}$$

{ V_T speed of tuning fork, $\ell_2 \rightarrow$ new length of pipe}.....(2)

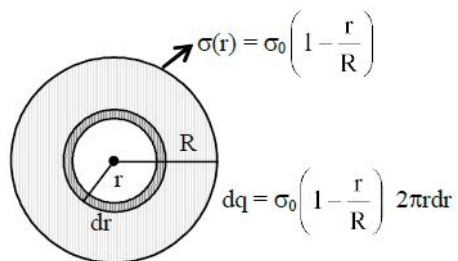
(1) $\div$ (2)

$$\frac{\ell_2 - \ell_1}{\ell_1} = \frac{-V_T}{V}$$

$$\frac{\ell_2 - \ell_1}{\ell_1} \times 100 = \frac{-2}{320} \times 100 = \frac{5}{8}$$

13. 5

Sol. $\phi_0 = \frac{\int dq}{\epsilon_0} = \frac{\int_0^R \sigma_0 \left(1 - \frac{r}{R}\right) 2\pi r dr}{\epsilon_0}$



$$\phi = \frac{\int dq}{\epsilon_0} = \frac{\int_0^{R/4} \sigma_0 \left(1 - \frac{r}{R}\right) 2\pi r dr}{\epsilon_0}$$

$$\therefore \frac{\phi_0}{\phi} = \frac{\sigma_0 2\pi \int_0^R \left(1 - \frac{r^2}{R}\right) dr}{\sigma_0 2\pi \int_0^{R/4} \left(r - \frac{r^2}{R}\right) dr}$$

$$= \frac{\frac{R^2}{2} - \frac{R^2}{3}}{\frac{R^2}{32} - \frac{R^2}{3 \times 64}} = \frac{32}{5}$$

Section – C

14. 0.40
15. 2.00
16. 1.00
17. 1.50
18. 12.50
19. 0.00

Chemistry**PART – II****Section – A**

20. A

21. B

Sol. $\Delta G^\circ = -RT \ln K_p$

22. B

Sol. Group – 3 cations $\rightarrow Cr^{+3}, Fe^{+3}, Al^{+3}$ Reagent $\rightarrow NH_4Cl/NH_4OH$

23. B

Sol. Meq of $KMnO_4$ = Meq. Of H_2O_2

$$16 \times 0.1 \times 5 = 24 \times x \times 2$$

$$x = \frac{16 \times 0.1 \times 5}{24 \times 2} = 0.167M$$

$$\text{Volume strength} = 11.2 \times \text{Molarity} \Rightarrow 11.35 \times 0.167 = 1.89$$

24. ABD

Sol. $r = r_0 \frac{n^2}{Z}$ As Z increases, r_0 decreases

Hence upon increasing Z, nuclear distance of node decreases.

25. C

26. BCD

Sol. The slag formed is $CaSiO_3$ which is lighter and has less melting point.

27. ABC

Sol. Kiliani Fischer method is for up gradation of C – chain.

28. BCD

Sol. At Boyle's Temperature; gas behaves ideally for a range of pressure.

29. ABCD

Section – B

30. 51

Sol. For $XeF_2 \Rightarrow Xe : F_2 = 2 : 1$ For $XeF_4 \Rightarrow Xe : F_2 = 2 : 10$ For $XeF_6 \Rightarrow Xe : F_2 = 2 : 40$

$$\text{Sum} = 1 + 10 + 40 = 51$$

31. 14

Sol. Borax $= Na_2B_4O_7 \cdot 10H_2O$ Kernite $= Na_2B_4O_7 \cdot 4H_2O$

32. 8

Sol. $0.8 = \left(1 - \frac{\alpha}{2}\right) \times 5 \times \frac{0.2/60}{0.02}$

$\alpha = 0.08$

Section – C

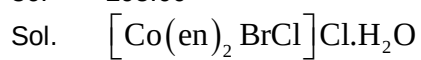
33. 8.00

Sol. G is an amide

34. 2.00

35. 2.00

36. 295.00



37. 3.00

38. 13.00

Sol. Nitrogen & halogens are more EN than boron. Number of H_2 molecular required saturating the compound.

Mathematics**PART – III****Section – A**

39. C

Sol. $\left(\frac{z_2}{z_1}\right)^2 = \left(\frac{OQ}{OP}\right)^2 e^{i2\theta}, \quad \frac{z_3}{z_1} = \frac{OR}{OP} e^{i2\theta}$

$$\Rightarrow \frac{z_2^2}{z_1 z_3} = \frac{\cos 2\theta}{\cos 2\theta} = \frac{2 + \sqrt{3}}{2\sqrt{3}}$$

$$\Rightarrow \cos 2\theta = \frac{\sqrt{3}}{2}.$$

$$\Rightarrow \theta = 15^\circ.$$

40. B

Sol. $f(x) = ax + b$

$$f(0) = 0 \Rightarrow b = 0.$$

41. B

Sol. $= 3(\hat{a} \cdot \hat{a} + \hat{b} \cdot \hat{b} + \hat{c} \cdot \hat{c}) - 2(\hat{a} \cdot \hat{b} + \hat{b} \cdot \hat{c} + \hat{c} \cdot \hat{a})$

$$\text{max is } 3 \times 3 + 2 \cdot \frac{3}{2} = 12.$$

42. C

Sol. $\alpha + \beta = -\frac{2h}{a}$

$$\alpha\beta = \frac{b}{a}$$

$$\gamma + \delta = -\frac{2h_1}{a_1}$$

$$\gamma\delta = \frac{b_1}{a_1}$$

$$\text{given } \gamma = \frac{\lambda\beta + \alpha}{\lambda + 1} \Rightarrow \lambda = \frac{\gamma - \alpha}{\beta - \gamma}$$

$$\text{similarly } \mu = \frac{\delta - \alpha}{\beta - \delta}.$$

$$\text{Now } \lambda + \mu = \frac{(\gamma - \alpha)(\beta - \delta) + (\beta - \gamma)(\delta - \alpha)}{(\beta - \gamma)(\beta - \delta)}$$

$$= \frac{\frac{2}{aa_1}(2hh_1 - a_1b - ab_1)}{(\beta - \gamma)(\beta - \delta)} = 0.$$

43. B, D

Sol. The sine curve chord going two points on the curve lies below the curve, hence (A) cannot be true. In log curve also, the same pattern is following but in tan curve and x^2 curves opposite pattern is followed.

44. A, C

 Sol. The 1st elements of $S_1, S_2, S_3, S_4, \dots$ are $\frac{1^2-1}{1}, \frac{2^2-1}{2}, \frac{3^2-1}{3}, \dots$
 $\Rightarrow S_i$ must be a set starting at $\frac{i^2-1}{i}$

$$S_{20} = \left\{ \frac{399}{20}, \frac{399+20}{20}, \frac{399+40}{20}, \dots, 20 \text{ term} \right\}$$

 $\Rightarrow$ third term is $\frac{439}{20}$

$$S_{20} = \frac{20}{2} \left[\frac{2 \times 399}{20} + 19 \times 1 \right] = 589.$$

45. C, D

$$\text{Sol. } D = \begin{vmatrix} m-2 & 2m-5 & 0 \\ m-1 & m^2-7 & -5 \\ 1 & 1 & -1 \end{vmatrix} = 0$$

$$\Rightarrow (m-3)(m^2-m+2) = 0$$

 For $m = 3$, the lines become parallel.

46. A, C, D

 Sol. $f(x) = \int_{-1}^x |\sin \pi t - \cos \pi t| dt$ is an increasing function.

47. A, D

48. B, C, D

 Sol. $a+b+c+d > \{a+b+d, a+b+c, b+c+d, a+c+d\}$

$$\Rightarrow x > 1$$

$$\text{Let } \max\{a, b, c, d\} = a$$

$$\Rightarrow a+b+c > b+c+d; a+c+d > b+c+d$$

$$\Rightarrow \frac{b}{a+b+c} + \frac{c}{b+c+d} + \frac{d}{a+c+d} < \frac{b+c+d}{b+c+d}$$

$$\Rightarrow x - \frac{a}{a+b+d} < 1 \Rightarrow x < 1 + \frac{a}{a+b+d} < 2$$

 Hence $1 < x < 2$.

Section – B

49. 2

$$\text{Sol. } p(x) = \frac{1-x^{18}}{1+x}$$

 coefficient of y^2 in $\left(\frac{1-(y-1)^{18}}{y} \right)$ is coefficient of y^3 in $(1-y)^{18} = {}^{18}C_3 = k = 816$

$$\Rightarrow \left[\frac{k}{400} \right] = 2.$$

50. 5

51. 1

Sol. $f(2011) = (2 + 0 + 1 + 1)^2 = 16$

$$f^2(2011) = (1 + 6)^2 = 49$$

$$f^3(2011) = (4 + 9)^2 = 169$$

$$f^4(2011) = (1 + 6 + 9)^2 = 256$$

$$f^5(2011) = (2 + 5 + 6)^2 = 169$$

$$\Rightarrow f^{2n}(2011) = 256$$

$$f^{2n+1}(2011) = 169.$$

Section – C

52. 32.00

53. 1118.00

54. 120.00

55. 15.00

56. 2.00

57. 5.00